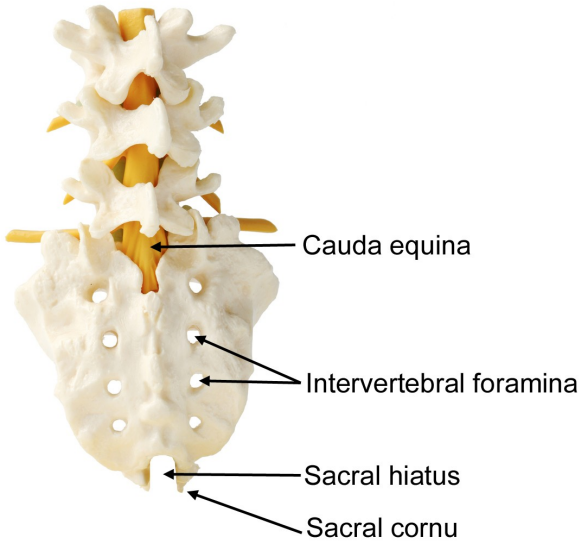
Caudal anesthesia is the most commonly used regional anesthetic in pediatric surgery. It is typically used as an adjunct to general anesthesia for surgery below the level of the umbilicus. It is useful in adults when the sacral spread of anesthesia is more important than the spread across lumbar dermatomes. It is also used increasingly in the management of chronic pain.

The procedure is performed in the lateral decubitus position in neonates and children because it provides easy access to the airway in anesthetized patients. In older children and adults, it is most commonly performed in the prone position, but lateral decubitus or knee-chest position can also be used.

A triangle can be drawn using the posterior superior iliac spines as the base with the apex of the triangle pointing inferiorly toward the skin folds of the buttocks. the sacral hiatus can usually be palpated directly under or adjacent to the apex of the triangle. The sacral cornua can be identified by palpating until the index finger or thumb sinks into the hollow space between them.

After sterile skin prep and a sterile drape is applied, the area of skin over the sacral hiatus is localized using a small-gauge 1.5-inch needle and 3-5 mL of1% lidocaine.

Sacral Anatomy



A 17-18 gauge Tuohy needle is inserted in the midline through the sacral hiatus. A 'pop' may be felt as the needle passes through the sacrococcygeal ligament. A syringe of air or saline is connected to the needle and a loss-of-resistance technique is used to determine that the needle has entered the epidural space. Bupivacaine 0.175-0.25% or ropivacaine 0.2% may be injected in a volume of 0.5-1 mL/kg. The safe maximal dose is 2.5 mg/kg. Epinephrine in a 1:200,000 concentration may be added to prolong the duration of the block and aid in earlier detection of intravascular injection. A 'whoosh' test can be performed by placing a stethoscope over the sacrum and listening for the rushing sound of 2-3 mL of injected air or local anesthetic passing through the epidural space.

The needle should never be inserted the full length of the shaft--it should stay below the S2 level to avoid puncturing the dura. If the needle swings easily from side-to-side at the hub, then needle has not been inserted correctly into the sacral canal. If cerebrospinal fluid is aspirated, the needle should be withdrawn and the injection should not be performed. If blood is aspirated, the needle should be withdrawn and the needle reinserted until no blood is aspirated. When injecting air using the loss-of-resistance technique, if a bulge is felt over the sacrum, then the needle most likely overlies the sacrum and should be reinserted.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 851.

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